

From Points to Structured Knowledge: Teaching HBIM Workflows in Architectural Heritage Documentation

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Abstract

The rapid digitisation of cultural heritage often prioritises visual fidelity and aesthetic appeal over semantic data integrity. Recent research and professional guidance increasingly frame Heritage Building Information Modelling not only as a documentation method, but also as a means of supporting conservation, maintenance, monitoring and intervention in built heritage (Arayici et al., 2017; Antonopoulou & Bryan, 2017). In the German context, however, established reference materials for architectural heritage documentation are still rooted in traditions of mainly analogue building survey based on 2D documentation, while standards and teaching resources for scan-to-(H)BIM workflows remain limited (Eckstein et al., 1999; Busen et al., 2015).

This study addresses this gap from an educational perspective by analysing how scan-to-(H)BIM teaching can train students in the AEC-related fields to transform survey data into structured, interoperable knowledge. The central research question is how to ensure that students not only master technical tools required for digitisation but also internalise the principles of interoperability and reusability during the complex and iterative process of architectural documentation (ICOMOS, 1964). While many academic programs focus primarily on the mechanics of laser scanning or photogrammetry, a significant gap remains in teaching the semantic enrichment phase (Garagnani, 2013; Kuroczyński et al., 2023). This transition is increasingly relevant for Linked Open Data, Web3D standards and FAIR-oriented heritage documentation.

The research is based on a longitudinal comparison of three consecutive editions of a heritage documentation course focused on places of worship, namely the Church of the Evangelical Parish of Mainz-Laubenheim in Germany, the former Synagogue in Münsterberg (nowadays in Ziębice, Poland) and the Orthodox Chapel of the Presentation of the Blessed Virgin Mary in Trześcianka (Poland). These case studies differed in access conditions, state of preservation and documentation challenges (Pepe et al., 2021; Sztwiertnia et al., 2019). Each edition followed a scan-to-(H)BIM workflow: on-site data

collection, point-cloud cleaning and registration, preparation of photographic and reference assets, and integration of the data into Archicad as the main BIM authoring environment. Students collaboratively developed HBIM models and organised object-based information through an IFC-oriented openBIM workflow. Rather than applying a uniform level of detail, students defined the modelling scope and information depth according to the historical, architectural and material significance of each element. This strategy encouraged them to distinguish between elements requiring precise geometric representation, elements requiring richer semantic description, and elements that could be represented schematically.

The study evaluates student workflows by identifying recurring modelling and information-structuring errors and comparing structured guidance with open-ended tasks. Outputs were assessed for clarity of representation levels, consistency of modelling decisions, completeness of IFC-based information, interoperability and future reuse, while feedback was collected through discussions after final presentations. The analysis indicates that structured guidance improved the consistency of IFC-based information, while open-ended tasks strengthened students' ability to justify modelling decisions but produced less uniform information structures. Drawing on ongoing research projects, including the HBIM Worms Synagogue case study (Kuroczyński & Argasiński, 2025), the course provides a basis for reusable educational materials and workflow guidelines for broader training formats, including blended-learning programmes and future summer schools, supporting capacity building and the development of young professionals in digital heritage documentation.

References

1. Arayici, Y., Counsell, J., Mahdjoubi, L., Nagy, G., Hawas, S., & Dweidar, K. (2017). *Heritage Building Information Modelling*. Routledge.
2. Antonopoulou, S., & Bryan, P. (2017). *BIM for Heritage: Developing a Historic Building Information Model*. Historic England.
3. Eckstein, G., Henes-Klaiber, U., Goer, M., & Baden-Württemberg. (1999). *Empfehlungen für Baudokumentationen: Bauaufnahme – Bauuntersuchung*. Stuttgart: Theiss.
4. Busen, T., Knechtel, M., Knobling, C., Nagel, E., Schuller, M., & Todt, B. (2015). *Bauaufnahme*. Edition MV-Wissenschaft.
5. ICOMOS. (1964). *The Venice Charter: International Charter for the Conservation and Restoration of Monuments and Sites*. Venice: ICOMOS.
6. Garagnani, S. (2013). *Building Information Modeling and real world knowledge: A methodological approach to accurate semantic documentation for the built environment*. Digital Heritage International Congress 2013.
7. Kuroczyński, P., Apollonio, F. I., Bajena, I. P., & Cazzaro, I. (2023). Scientific reference model: Defining standards, methodology and implementation of serious 3D models in archaeology, art and architectural history. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLVIII-M-2-2023, 895–902.
8. Pepe, M., Costantino, D., & Restuccia Garofalo, A. (2021). Scan-to-HBIM for conservation and preservation of cultural heritage building: The case study of San Nicola in Montedoro church (Italy). *Applied Geomatics*, 15, 607–621.
9. Sztwiertnia, D., Ochałek, A., Tama, A., & Lewińska, P. (2019). HBIM (Heritage Building Information Model) of the Wang Stave Church in Karpacz: Case study. *International Journal of Architectural Heritage*, 15(5), 713–727.
10. Kuroczyński, P., & Argasiński, K. (2025). Implementing HBIM for Cultural Heritage Preservation – From Documentation to Sustainable Management. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, X-M-2-2025, 175–184.